Ivan Ochoa Lynch
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Cheating, Beliefs, and Curved Grading as a Bayesian Game

In the classroom, academic cheating is often viewed as a failure of moral character; a sign of a weak moral compass or poor work ethic. While this perspective has some merit, it overlooks the strategic reasoning behind such decisions. Game theory raises a more challenging question: Why, and under what circumstances, does cheating become rational for students who would prefer to work honestly?

The answer depends largely on students' perceptions of and knowledge about their academic environment. If a student believes that almost no one cheats, peer cheating poses little relative disadvantage, though cheating may still offer a direct benefit. But if a student believes many classmates cheat, honesty may feel like unilateral disarmament, leaving them in a relatively weaker position than their dishonest peers, especially in an academic context where they are graded on a curve or compared to a median. The classroom then becomes a strategic environment in which a student's choices depend not only on their own actions but also on their beliefs about others' behavior.

This paper aims to model exam cheating as a Bayesian game with incomplete information. Students cannot directly observe the overall level of cheating in their class. Instead, they rely on noisy signals – rumors, unusual score distributions, whispered stories, and performance feedback – to shape their beliefs and actions. The paper makes a narrow claim: grade curves amplify the effect of beliefs about cheating, but they do not make cheating optimal for every student. Under absolute grading, peer cheating primarily affects the moral cost of cheating. Under curved or median-enforced grading, it also affects students' relative rankings and, in turn, their grades. As grading becomes more relative, believing that others are cheating raises the cost of remaining honest.

Consider a class with n students. Each student chooses to either complete honest work, H, or to cheat, C. Student i has belief p_i, interpreted as the expected share of other students who cheat, or equivalently the probability that a randomly selected classmate cheats. The student also has a moral type $\eta_i \sim G[0, \bar{\eta}]$, where a higher η_i means cheating carries a higher private moral or psychological cost. Other students know the distribution G, but not student i's realized type. The game is Bayesian because students differ in a private-payoff-relevant type and because students choose simultaneously after observing their own type. For the symmetric equilibrium calculation, assume independent draws from the same continuous distribution G, with a positive finite upper bound. Assume other parameters are common knowledge.

Let B be the baseline payoff from taking the exam honestly. Let $g > 0$ be the private benefit from cheating, such as saved study time, copied answers, or unauthorized help. Let r be the probability of detection and F be the penalty if the student is caught, such that rF is the expected formal punishment. Let $\lambda \geq 0$ measure the strength of relative grading. When $\lambda = 0$

the class is graded absolutely, whereas when λ is high, grades are more tournament-like. Finally, let $d > 0$ be the relative disadvantage from staying honest when others cheat. There are at least two students; beliefs and detection probabilities lie between zero and one, and penalties are nonnegative. All payoff terms use the same utility units. The expected punishment is additive, with the penalty including any loss of the cheating benefit upon detection. These are interim expected payoffs: the linear peer terms can be obtained by using the realized share of other students who cheat and then taking expectations. The model assumes cheating avoids the relative loss imposed on honest work. It does not derive this asymmetry from a particular grading formula.

The expected honest payoff is:

$$U_i(H) = B - \lambda p_i d$$

This says that an honest student gets the baseline payoff B, but suffers an expected relative loss when other students cheat, and the class is curved. If $\lambda = 0$, that relative loss disappears, presupposing there are no other measures that weigh students against each other. Peer cheating may still feel unfair, but it does not mechanically lower an honest student's grade.

The expected cheating payoff is:

$$U_i(C) = B + g - rF - (\eta_i - \alpha p_i)$$

The term $\eta_i - \alpha p_i$ is the behavioral part of the model. The student has an individual moral cost of η_i, but that cost may fall when cheating seems common. If a student believes cheating is rare, cheating feels like a personal violation. If the student believes cheating is normal, the same act may feel defensive. The parameter $\alpha \geq 0$ measures how strongly perceived peer cheating lowers moral cost. This linear term can become negative, so the model allows a net psychological reward when cheating seems common. Keeping moral cost nonnegative would require a different payoff and cutoff.

Cheating is a best response when:

$$B + g - rF - \eta_i + \alpha p_i \geq B - \lambda p_i d$$

Canceling B this becomes,

$$g + p_i(\lambda d + \alpha) \geq rF + \eta_i$$

The left side represents the benefit of cheating. It includes the direct private benefit, g, the relative protection from cheating when others cheat, $\lambda p_i d$, and the social normalization effect αp_i. The right side represents expected punishment, rF, plus the student's personal moral resistance, η_i. The cutoff belief is:

$$p_i^* = \frac{rF+\eta_i-g}{\lambda d+\alpha}$$

Student i cheats if $p_i \geq p_i^*$. This cutoff interpretation assumes $\lambda d + \alpha > 0$. If $\lambda d + \alpha = 0$, beliefs do not affect the cheating decision, and the student only cheats when $g \geq rF + \eta_i$. If $p_i^* < 0$, the student cheats even when perceived cheating is zero; if $p_i^* > 1$, the student does not cheat for any feasible belief. At the cutoff, the student is indifferent; I use cheating as the tie rule. A cutoff of zero allows cheating at every feasible belief, while a cutoff of one allows it only at certainty. This is the core result. Cheating does not require that the student be amoral. For a student with an interior cutoff, perceived peer cheating must be high enough that honesty is too costly relative to cheating.

The non-curved case clarifies the role of λ. If $\lambda = 0$, the condition becomes:

$$g + \alpha p_i \geq rF + \eta_i.$$

Peer cheating matters only because it lowers the moral cost of cheating. The student may think "everyone does this," but the grade formula itself does not punish honesty. By contrast, in a curved class, $\lambda > 0$, peer cheating has a material effect. The student may think, "If others inflate their scores and the median is fixed, my honest score is worth less." The curve changes the strategic meaning of cheating by making grades more relative.

Kajackaite and Gneezy's "Incentives and Cheating," published in *Games and Economic Behavior,* helps justify the moral-cost part of this model. They distinguish between people with infinite lying costs, people with zero lying costs, and people with finite positive lying costs. The last group cheats when the benefit exceeds the cost. Their experiment also shows that exposure concerns matter: when fear of being exposed is reduced in a modified "mind" game, lying rises with incentives (Kajackaite and Gneezy 2017). In the model above, that maps naturally onto η_i , g, and rF. Their exposure concern is broader than formal punishment, so this is an analogy rather than a direct measure of the expected penalty. Some students do not cheat because cheating has a private cost. Others refrain because detection is possible. Raise the benefit or lower the perceived cost, and some types switch actions.

Benistant, Galeotti, and Villeval add the social-information channel. In their dynamic die-reporting experiment, they concluded that subjects lie more in competitive settings than in non-competitive ones when both players can cheat. They also find that continuous feedback on another participant's reports increases dishonesty under piece-rate incentives, but not under tournaments (Benistant et al. 2021). For this paper, these findings motivate a possible

information channel; they do not establish that tournament feedback always increases cheating. In the model, feedback moves p_i. If p_i crosses p^*, the student's best response changes.

Martinelli, Parker, Pérez-Gea, and Rodrigo bring the model closer to the classroom. They studied a policy experiment in Mexican high schools where monetary incentives were tied to standardized math exams. Their evidence suggests that cheating was more common when students received incentives, and that cheating increased over time in treatments with student incentives (Martinelli et al. 2018). This is consistent with learning to cheat, but does not by itself identify a change in beliefs or moral cost. Students do not choose in a vacuum. If cheating appears to spread, later students may update p_i upward and face a lower moral barrier because cheating has become more familiar.

To make the Bayesian equilibrium explicit, use the independent moral types defined above. The formal calculation uses a common belief and does not specify private signals. Suppose η_i is drawn from a distribution with CDF G on $[0, \bar{\eta}]$. Given the common belief p, shared by all students, a student cheats when:

$$\eta_i \leq g + p(\lambda d + \alpha) - rF$$

Extend the CDF to equal zero below the type support and one above it. The expected share of cheaters is therefore:

$$q(p) = G(g + p(\lambda d + \alpha) - rF)$$

Under these assumptions, a symmetric Bayesian Nash equilibrium is characterized by correct belief $p = q(p)$.

In words, the perceived cheating rate must equal the expected cheating rate generated by students' cutoff strategies. In a finite class, the realized share can differ from that expectation. Continuity guarantees at least one equilibrium. If G has a density, a sufficient condition for uniqueness is that the combined peer effect times the density stays below one throughout the relevant range. Multiple equilibria are possible, but a strong peer effect alone does not guarantee them. If students expect little cheating, only low-moral-cost types cheat so that the low-cheating belief can be confirmed. If students expect widespread cheating, more types cross the cutoff, so the high-cheating belief can also be confirmed. For example, with moral types uniform on [0,1], a direct benefit net of expected punishment of -0.25, and a combined peer effect of 1.5, cheating rates of zero, one-half, and one are all equilibria. These outcomes can occur under the same formal rules and information structure; different self-consistent expectations support them.

The model does not establish that better information reduces cheating or volatility. If students observed peer actions before choosing, the timing of the game would change, and p_i could describe an observed share rather than an expectation. In real exam settings, students see only noisy signals. A rumor, a suspicious grade jump, or a friend's confession can matter because they change posterior beliefs. Bayesian updating would require a prior and signal likelihoods,

which are not specified here. With a nondegenerate prior, a signal raises the posterior probability of widespread cheating if it is more likely under widespread cheating than under low cheating. The point is not that the students explicitly calculate posteriors. The point is that incomplete information makes beliefs part of the payoff environment.

The model also gives clean comparative statics in the interior case, where $0 < p^* < 1$ and $rF + \eta_i - g > 0$. Stronger relative grading lowers the cutoff because it increases $\lambda d + \alpha$. Formally,

$$\frac{\partial p_i^*}{\partial \lambda} = \left(-\frac{d(rF+\eta_i-g)}{(\lambda d+\alpha)^2}\right) < 0.$$

Holding the other parameters fixed, higher detection probability or harsher penalties weakly raise p^* because rF cannot fall, and rises when the other component is positive. A stronger moral type raises p^*_i. More normalization of cheating lowers p^* through α. These are individual-cutoff comparisons, not equilibrium derivatives; equilibrium changes also depend on how beliefs adjust. Feedback can push the student either way. If feedback convinces students that cheating is rare, p_i falls, and honesty becomes easier to sustain. If feedback convinces them that cheating is common, the same feedback can destabilize honesty.

The enforcement implication is that penalties alone do not target the belief channel. Raising penalties is one tool, but it is not the whole mechanism. Within this model, expected punishment above the direct benefit plus both peer effects makes honesty strictly optimal for every type and every feasible belief. A policy that raises F while allowing students to believe cheating is widespread may still fail in a curved class. A policy that lowers p_i, through visible monitoring, clear proctoring, or credible communication that cheating is uncommon, can affect perceived cheating, but any equilibrium belief must still agree with the behavior it generates. The model does not establish that each of these policies lowers beliefs. The goal should not just be deterrence but also to keep honest students from believing they are the last ones still following the rules.

Curves can be defensible. They may correct for exam difficulty and make grades comparable across sections, but they also make beliefs about cheating more strategically important. In an absolute-grading class, another student's cheating is mostly an integrity problem. In a curved or median-enforced class, it is also a rank problem. The issue is that perceived dishonesty can change best responses. A classroom fails strategically when enough students come to believe that honesty is no longer a safe option.

References

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